

# Joe Benton

## EDUCATION

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- D.Phil. in Statistics** 2021 – 2023  
*Department of Statistics, University of Oxford*  
Thesis title: *Generative Modelling: Theory and Applications*, supervised by Arnaud Doucet and George Deligiannidis
- B.A. with M.Math. in Mathematics** 2017 – 2021  
*Trinity College, University of Cambridge*  
Graduated with Distinction (6th out of c. 250 in year)  
Thesis title: *Activated Random Walks*, supervised by Perla Sousi

## PROFESSIONAL EXPERIENCE

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- Member of Technical Staff (Manager)** | *Anthropic* July 2025 – Present  
Managed the Scalable Oversight team, working on providing reliable training rewards without human supervision.  
Research lead for the *Anthropic Fellows Program*, a 4–6 month fellowship program for alignment research.
- Member of Technical Staff** | *Anthropic* November 2023 – July 2025  
Worked on training model organisms of misalignment from reward hacking, chain-of-thought monitoring and evaluations, threat modeling and control evaluations for the Responsible Scaling Policy, and ASL-3 adversarial robustness.
- Technical Staff** | *UK Frontier AI Taskforce* Fall 2023  
Part of the founding technical team of the UK Frontier AI Taskforce (now UK AI Security Institute). Produced technical demonstrations of AI risk for presentation in front of world leaders at the first AI Safety Summit.
- Machine Learning Researcher** | *Redwood Research* Winter 2022 – 2023  
Developed and studied causality and fine-tuning based interpretability methods, with applications to mechanistic anomaly detection. Supervised an internship project aiming to automate interpretability techniques.

## VOLUNTEERING

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- Research Mentor** | *Anthropic Fellows Program, ML Alignment & Theory Scholars, SPAR* Fall 2022 – Present  
Supervised 15+ fellows, with publications in AI control evaluations, CoT monitoring and faithfulness, scalable oversight, automated alignment research, and interpretability.
- Trustee, Cofounder** | *Raise: A Celebration of Giving* 2018 – 2025  
Trustee and co-founder of Raise, a student charity initiative raising over £500,000 for the *Against Malaria Foundation*.

## AWARDS

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- International Mathematics Olympiad (1 Gold – 7th out of 615, 3 Silver)** 2014 – 2017
- International Olympiad in Informatics (1 Gold – 6th out of 304, 1 Silver, 1 Bronze)** 2015 – 2017
- Romanian Masters in Mathematics (3 Gold – 2nd, 5th and 6th overall)** 2015 – 2017

## INVITED TALKS

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- University of Warwick Algorithms and Computationally Intensive Inference Seminars, 2023
- Royal Statistical Society International Conference, Probabilistic and Statistical Aspects of Machine Learning Discussion Meeting, 2023

## SERVICE

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- Program Chair, AAAI 2025 Alignment Track
- Reviewed for NeurIPS, ICML, ICLR, JMLR, TMLR, JRSS-B, NeurIPS 2023 SoLaR Workshop, NeurIPS 2023 Workshop on Diffusion Models, ICML 2023 Workshop Frontiers4LCD, Cooperative AI Foundation

- Natural Emergent Misalignment from Reward Hacking in Production RL*. Monte MacDiarmid, Benjamin Wright, Jonathan Uesato, **Joe Benton**, Jon Kutasov, Sara Price et al. *arXiv preprint arXiv:2511.18397*, 2025.
- Optimizing AI Agent Attacks With Synthetic Data*. Chloe Loughridge, Paul Colognese, Avery Griffin, Tyler Tracy, Jon Kutasov, **Joe Benton**. *arXiv preprint arXiv:2511.02823*, 2025.
- Evaluating Control Protocols for Untrusted AI Agents*. Jon Kutasov, Chloe Loughridge, Yuqi Sun, Henry Sleight, Buck Shlegeris, Tyler Tracy, **Joe Benton**. *arXiv preprint arXiv:2511.02997*, 2025.
- SHADE-Arena: Evaluating Sabotage and Monitoring in LLM Agents*. Jonathan Kutasov, Yuqi Sun, Paul Colognese, Teun van der Weij, Linda Petrini, Chen Bo Calvin Zhang, John Hughes, Xiang Deng, Henry Sleight, Tyler Tracy, Buck Shlegeris, **Joe Benton**. *arXiv preprint arXiv:2506.15740*, 2025.
- Reasoning Models Don't Always Say What They Think*. Yanda Chen, **Joe Benton**, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, Ethan Perez. *arXiv preprint arXiv:2505.05410*, 2025.
- Inverse Scaling in Test-Time Compute*. Aryo Pradipta Gema, Alexander Hägele, Runjin Chen, Andy Arditi, Jacob Goldman-Wetzler, Kit Fraser-Taliente, Henry Sleight, Linda Petrini, Julian Michael, Beatrice Alex, Pasquale Minervini, Yanda Chen, **Joe Benton**, Ethan Perez. *arXiv preprint arXiv:2507.14417*, 2025.
- Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety*. Tomek Korbak, Mikita Balesni, Elizabeth Barnes, Yoshua Bengio, **Joe Benton**, Joseph Bloom, Mark Chen, Alan Cooney, Allan Dafoe, Anca Dragan, Scott Emmons, Owain Evans, David Farhi, Dan Hendrycks, et al. *arXiv preprint arXiv:2507.11473*, 2025.
- Teaching Models to Verbalize Reward Hacking in Chain-of-Thought Reasoning*. Miles Turpin, Andy Arditi, Meihua Li, **Joe Benton**, Julian Michael. *arXiv preprint arXiv:2506.22777*, 2025.
- Constitutional Classifiers: Defending against Universal Jailbreaks across Thousands of Hours of Red Teaming*. Mrinank Sharma, Meg Tong, Jesse Mu, Jerry Wei, Jorrit Kruthoff, Scott Goodfriend, Euan Ong, Alwin Peng, Raj Agarwal, Cem Anil, Amanda Askell, Nathan Bailey, **Joe Benton**, Emma Bluemke, et al. *arXiv preprint arXiv:2501.18837*, 2025.
- Sabotage Evaluations for Frontier Models*. **Joe Benton**, Misha Wagner, Eric Christiansen, Cem Anil, Ethan Perez, Jai Srivastav, Esin Durmus, Deep Ganguli, Shauna Kravec, Buck Shlegeris et al. *arXiv preprint arXiv:2410.21514*, 2024.
- Many-shot Jailbreaking*. Cem Anil, Esin Durmus, Mrinank Sharma, **Joe Benton**, Sandipan Kundu, Joshua Batson, Nina Rimskey, Meg Tong, Jesse Mu, Daniel Ford et al. *Advances in Neural Information Processing Systems*, 2024.
- When Do Universal Image Jailbreaks Transfer Between Vision-Language Models?*. Rylan Schaeffer, Dan Valentine, Luke Bailey, James Chua, Cristóbal Eyzaguirre, Zane Durante, **Joe Benton**, Brando Miranda, Henry Sleight, John Hughes et al. *NeurIPS 2024 Workshop on Responsibly Building the Next Generation of Multimodal Foundational Models*, 2024.
- From Denoising Diffusions to Denoising Markov Models*. **Joe Benton**, Yuyang Shi, Valentin De Bortoli, George Deligiannidis, Arnaud Doucet. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(2):286–301, 2024.
- Nearly  $d$ -Linear Convergence Bounds for Diffusion Models via Stochastic Localization*. **Joe Benton**, Valentin De Bortoli, Arnaud Doucet, George Deligiannidis. *International Conference on Learning Representations*, 2024.
- Error Bounds for Flow Matching Methods*. **Joe Benton**, George Deligiannidis, Arnaud Doucet. *Transactions on Machine Learning Research*, February 2024.
- Alpha-divergence Variational Inference Meets Importance Weighted Auto-Encoders: Methodology and Asymptotics*. Kamélia Daudel, **Joe Benton**, Yuyang Shi, Arnaud Doucet. *Journal of Machine Learning Research*, 24(243):1–83, 2023.
- Measuring Feature Sparsity in Language Models*. Mingyang Deng, Lucas Tao, **Joe Benton**. *NeurIPS 2023 Workshop on Socially Responsible Language Modelling Research*, 2023.
- A Continuous Time Framework for Discrete Denoising Models*. Andrew Campbell, **Joe Benton**, Valentin De Bortoli, Tom Rainforth, George Deligiannidis, Arnaud Doucet. *Advances in Neural Information Processing Systems*, 2022.
- Polysemanticity and Capacity in Neural Networks*. Adam Scherlis, Kshitij Sachan, Adam S. Jermyn, **Joe Benton**, Buck Shlegeris. *arXiv preprint, arXiv:2210.01892*, 2022.

## RESEARCH BLOG POSTS

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*A3: An Automated Alignment Agent for Safety Finetuning.* Jifan Zhang, Henry Sleight, **Joe Benton**. *Anthropic Alignment Science Blog*, 2026.

*Towards Training-Time Mitigations for Alignment Faking in RL.* Johannes Gasteiger, Vlad Mikulik, Hoagy Cunningham, Misha Wagner, Benjamin Wright, Jonathan Uesato, **Joe Benton**, Monte MacDiarmid, Fabien Roger, Evan Hubinger. *Anthropic Alignment Science Blog*, 2025.

*Training Fails to Elicit Subtle Reasoning in Current Language Models.* Misha Wagner, Fabien Roger, Hoagy Cunningham, Johannes Gasteiger, **Joe Benton**, Vlad Mikulik. *Anthropic Alignment Science Blog*, 2025.